



USA TRIATHLON EDUCATION LT1, LT2 and VO2

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Overview

LT1, LT2, and VO2 max are the three physiological reference points that matter most for endurance athletes. If you know where they sit, you can structure training and testing for almost any athlete and build any zone system from them.

For short-course and draft-legal athletes, all three matter because racing under an hour leans on VO2 work. For long-course athletes, LT1 and LT2 usually carry most of the load because far less training happens at the high end.

The physiology is the same in every sport. Only the names change: LT2 is FTP on the bike, Critical Swim Speed in the pool, anaerobic threshold or lactate threshold on the run, and OBLA in the lab.

LT1: the aerobic threshold

LT1 is the point where lactate first rises above baseline. It is also called the aerobic threshold or Fat Max, and it is typically placed around 2 mmol/L.

- **Relationship to LT2:** about 80% of LT2 for elite athletes and about 70% for age groupers.
- **What limits you here:** fueling and muscular endurance, not cardiovascular capacity. With enough fuel it is theoretically sustainable indefinitely.
- **Who lives near it:** Ironman athletes typically race at or just above LT1.
- **Training use:** this is where much of the long, slow aerobic base work happens.
- **Identification:** hard to spot without lactate or gas-exchange testing. Most training software estimates it from a known LT2 value.

Baseline lactate itself runs roughly 1.0 to 1.3 mmol/L in trained endurance athletes and about 1.6 to 2.0 in untrained people, so LT1 is better read as the first sustained rise than as a fixed number.

LT2: the anaerobic threshold

LT2 is the point where lactate accumulates faster than the body can clear it, typically around 4 mmol/L (about 3.8 to 4.1). It is the most important reference point for endurance athletes, especially in middle-distance racing.

- **Sustain time:** roughly 30 to 70 minutes depending on genetics and training. It is commonly framed as the highest intensity an athlete can hold for 45 to 60 minutes.
- **Trainability:** highly trainable. This is the point at which we see a rapid increase in anaerobic contributions to an athlete's output.

- **Why breathing jumps:** blood turns slightly more acidic, so the body breathes off CO₂ to rebalance pH. We breathe to eliminate CO₂, not to get oxygen, which makes heavy breathing a clear threshold signal.
- **Heart rate deflection:** at LT2 the rise in heart rate slows even as power or pace keeps climbing, because energy production shifts more toward anaerobic metabolism. Plot heart rate against power or pace from a hard progressive effort or race to find it.
- **Field cues:** sudden increase in respiratory rate, flushed cheeks, trouble speaking full sentences, and RPE around 7 out of 10 (13 to 14 on the Borg 6 to 20 scale), described as hard but sustainable.

VO₂ max

VO₂ max is the maximum volume of oxygen the body can use per unit of body weight. It matters most for athletes racing under an hour, particularly short-course athletes.

- **Its limits as a predictor:** above roughly 70 to 75 mL/kg/min it stops being the main performance predictor. Lactate dynamics and Functional Reserve Capacity (FRC) become the limiter.
- **Higher is not automatically faster:** lactate tolerance and response correlate more strongly with race outcomes than VO₂ alone.
- **Trainability:** VO₂ can improve, with gains of up to 20% observed over an athlete's career.
- **Scale:** elite values are around 97 to 101 mL/kg/min, as in the cases of Tadej Pogačar (about 97) and Christian Blumenfeld (about 101).

How the three markers shape zones and athlete profiles

LT1 and LT2 set the borders between zone groups, and VO₂ anchors the top zones. The gap between LT1 and LT2 and the shape of the lactate curve tell you what kind of athlete you have.

Feature	Long-course profile	Short-course profile
Curve shape	Flatter for longer, LT1 and LT2 close together	Steeper, earlier rise, bigger gap between LT1 and LT2
Anaerobic system	Underdeveloped, low FRC	Developed, higher FRC
Peak lactate	About 8 to 13 mmol/L	About 15 to 25 mmol/L
Typical origin	High-volume, low-intensity training	Intensity-focused development

An athlete with a higher LT2 but low FRC can be dropped in a variable-pace race by a rival with a lower LT2 and more FRC. It is also easier to build an aerobic base in an anaerobic athlete than to build anaerobic capacity in an aerobic one, which is why athletes move from short course to long course far more often than the reverse.

Testing options and coaching notes

Lactate step test. Use 3.5 to 4 minute steps for newer athletes and 4.5 to 6 minutes for advanced athletes, so lactate has time to stabilize. Very long protocols of 8 or 9 steps create fatigue that skews

the top end. Record power, pace, and heart rate at each finger prick, then read LT1 at the first rise, LT2 at the sharp increase, and maximum tolerance at the end.

Without lactate. Use 10-watt jumps every 2 minutes and find the deflection point mathematically from heart rate or power response. You can also rely on respiratory rate, RPE around 6.5 to 7, or known race pace data.

Frequency. Full testing every 3 to 6 months suits most athletes, and annually for highly trained athletes whose physiology is stable. Spot checks every 4 to 8 weeks confirm zone adherence. Retest early after a major shift in training focus or when threshold cues change.

Watch for shifts in the field. If an athlete who held threshold at 6:00 per mile now shows threshold breathing at 5:45, fitness has likely moved and it is time to retest.

Test in the first week after an adaptation week, when the athlete is fresh.